RetailOS

Software Design Specification

Version 1.0

Multi-Tenant SaaS Platform
for Retail Management in Algeria

Prepared by :	RetailOS Team
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This document is the single source of truth for the RetailOS platform. All architectural decisions, business requirements, and implementation specifications derive from this document.

Document Control

Revision History

Version	Date	Author	Changes
1.0	2026-07-07	RetailOS Team	Initial release

Distribution List

Recipient	Role
Project Lead	Architecture decisions
Development Team	Implementation reference
QA Team	Testing requirements
Stakeholders	Business requirements verification

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Part I — Business Analysis

Chapitre 1

Vision

1.1 Executive Summary

RetailOS is a multi-tenant SaaS platform designed to be the **Retail Operating System** for Algeria. It combines Enterprise Resource Planning (ERP), Artificial Intelligence (AI), Business Intelligence (BI), and Decision Support Systems (DSS) into a single cloud platform.

The first vertical is **supermarket management**. The architecture is deliberately designed to be extensible to other retail verticals — pharmacies, hardware stores, electronics shops, wholesalers, clothing stores, and bookstores — with minimal changes to the core architecture.

1.2 Core Differentiator

The core differentiator of RetailOS is not basic transaction recording — every competitor does that. RetailOS transforms operational data into **actionable business recommendations** using operations research and decision support methodologies.

The POS, inventory, purchasing, finance, and CRM modules become the **data collection layer**. The true competitive advantage is the optimization engine that transforms this operational data into recommendations : optimal purchase quantities, supplier rankings using AHP/TOPSIS/PROMETHEE, demand forecasting, dynamic pricing, waste reduction, staff scheduling, and scenario simulation.

RetailOS does not just digitize operations. It **augments managerial intelligence**.

1.3 Strategic Positioning

- **Not a simple POS :** RetailOS is a full retail management platform with AI-powered decision support.
- **Not just for supermarkets :** The architecture supports multiple retail verticals. Supermarkets are the first implementation.
- **Not just for Algeria :** While the initial target market is Algeria with full DÉCRET 05-468 compliance, the architecture supports internationalization.
- **Purpose-built for Algeria :** Native DZD support, French/Arabic/English localization, Algerian fiscal compliance, offline capability for unreliable connectivity.

1.4 Target Audience

RetailOS targets small and medium-sized retail businesses in Algeria that currently rely on desktop software, spreadsheets, or manual processes. The platform is designed to be accessible to non-technical users while providing enterprise-grade features for growing businesses.

Chapitre 2

Market Analysis

2.1 Introduction

The Algerian retail sector has experienced significant growth during the last decade. Supermarkets, minimarkets, wholesalers, and neighborhood grocery stores increasingly require digital tools to improve inventory management, sales tracking, purchasing decisions, and financial reporting. Despite this growth, a considerable proportion of retailers continue to rely on traditional desktop software, spreadsheets, or manual processes.

2.2 Current Challenges

Retail businesses in Algeria commonly encounter the following operational challenges :

- **Frequent stock shortages** — products run out unexpectedly, leading to lost sales and customer dissatisfaction.
- **Overstocking** — excess inventory ties up capital and increases waste, especially for perishable goods.
- **Product expiration and waste** — poor demand forecasting leads to products expiring before sale.
- **Manual purchasing decisions** — managers order based on intuition rather than data-driven analysis.
- **Lack of sales forecasting** — no ability to predict demand spikes (Ramadan, holidays, seasonal patterns).
- **Limited reporting capabilities** — existing tools produce basic reports without actionable insights.
- **Absence of decision support** — managers have no tool to evaluate suppliers objectively or simulate scenarios.
- **Poor supplier evaluation** — supplier selection is based on price alone, ignoring delivery, quality, and reliability.
- **Manual inventory counting** — time-consuming, error-prone, and infrequent.
- **Limited integration between departments** — sales, inventory, purchasing, and finance operate in silos.

2.3 Digital Transformation Drivers

Retail digitalization in Algeria is accelerating due to several factors :

- **Growing competition** — new supermarkets and modern retail formats are entering the market.

- **Better internet accessibility** — 3G/4G coverage is expanding, though reliability varies outside major cities.
- **Affordable cloud computing** — makes SaaS economically viable for SMBs.
- **Mobile device proliferation** — smartphones are nearly universal among business owners and managers.
- **Barcode technology** — affordable barcode printers and scanners are now accessible to small businesses.
- **Artificial Intelligence** — cloud-based AI services are becoming affordable for non-enterprise users.
- **Data-driven management** — a new generation of managers expects analytics and insights, not just data recording.

2.4 Competitive Analysis

RetailOS occupies a unique position : it combines full Algerian compliance with AI-powered decision support at a price point accessible to SMBs.

Competitor	**Description**	**AI/Optimization**	**Algerian Compliance**	**SaaS**
eFawtara	Online invoicing platform	No	Partial (invoices only)	Yes
QuickBooks	Global accounting software	No	No	Yes
Local Desktop POS	Legacy POS solutions	No	Partial (no updates)	No
SAP/Oracle	Enterprise ERP	Yes	No (not adapted)	Yes (expensive)
Odoo	Open-source modular ERP	Limited	No (requires customization)	Yes (self-hosted)
RetailOS	**Retail OS for Algeria**	**Yes (MCDA, forecasting)**	**Full**	**Yes**

Table 2.1 – Competitive Landscape

Chapitre 3

Problem Statement

3.1 The Gap in Existing Solutions

Most retail management systems currently available focus primarily on recording business transactions. While they provide essential operational functionality — product catalogs, sales recording, basic inventory tracking — they rarely assist managers in making informed strategic decisions.

3.2 Questions Existing Systems Cannot Answer

Existing systems generally do not answer questions that are critical to retail management :

- Which products should be reordered *today* ?
- Which supplier provides the best *overall value* (considering price, delivery time, reliability, quality) ?
- Which products are likely to *expire this week* ?
- How much inventory should be purchased *before Ramadan* ?
- Which promotion will *maximize profit* without eroding margins ?
- Which employee schedule *minimizes labor costs* while maintaining service quality ?
- What will happen to profitability if prices *increase by 5%* ?

3.3 The Consequence

Managers are required to make these decisions manually, relying primarily on experience and intuition rather than analytical evidence. This leads to suboptimal purchasing, excess waste, lost sales, and reduced profitability.

3.4 The RetailOS Proposition

RetailOS proposes an integrated Decision Support System capable of transforming operational data into actionable business recommendations. The platform does not replace the manager's judgment — it **augments** it with data-driven insights.

Chapitre 4

Project Objectives

4.1 General Objective

Develop an enterprise-grade cloud platform capable of managing all operational aspects of retail businesses while integrating optimization, forecasting, artificial intelligence, and decision support.

4.2 Specific Objectives

The system shall address the following operational areas :

1. **Product Management** — Create, categorize, barcode, and manage product catalogs with full traceability.
2. **Inventory Management** — Track stock levels in real-time, manage stock movements, prevent shortages and overstocking.
3. **Warehouse Management** — Organize storage locations, manage transfers between warehouses and stores.
4. **Supplier Management** — Maintain supplier records, evaluate performance, rank using MCDA methods.
5. **Purchasing** — Generate purchase orders, track deliveries, optimize reorder quantities.
6. **Customer Management** — Track customer data, purchasing history, loyalty points.
7. **Employee Management** — Manage staff, roles, schedules, attendance, and commissions.
8. **Financial Management** — Compliant invoicing, expense tracking, tax reporting, profit and loss.
9. **Point of Sale** — Fast, reliable, offline-capable POS for daily sales operations.
10. **Business Intelligence** — Dashboards, reports, KPIs, and visual analytics.
11. **Demand Forecasting** — Predict future sales using historical data, seasonality, and external factors.
12. **Inventory Optimization** — Calculate optimal reorder points, safety stock, and economic order quantities.
13. **Waste Reduction** — Predict expirations, recommend markdowns, track waste metrics.
14. **Supplier Ranking** — Multi-criteria evaluation using AHP, TOPSIS, PROMETHEE methods.
15. **Scenario Simulation** — Model what-if scenarios (price changes, demand shifts, supplier changes).

16. **Algerian Compliance** — Full compliance with NIF, NIS, RC, AI, TVA requirements and DÉCRET 05-468.

Chapitre 5

Stakeholders

The platform serves several categories of users, each with distinct needs and permission levels.

5.1 Platform Administrator

Responsible for platform maintenance, billing, tenant management, monitoring, and technical administration. This role operates across all tenants and manages the SaaS infrastructure itself. The platform administrator has access to all tenants but should only access data for troubleshooting purposes.

5.2 Business Owner

Owns one or more retail stores. Responsible for strategic decisions, financial monitoring, and organizational management. Has full access to all modules within their tenant, including all reports, AI recommendations, and simulation results. Can create and manage user accounts for their organization.

5.3 Store Manager

Responsible for supervising day-to-day operations in one or more stores. Monitors inventory, purchasing, employees, and performance indicators. Can create purchase orders, manage stock transfers, and approve returns. Does not have access to platform-level settings or other tenants' data.

5.4 Cashier

Processes customer sales through the Point-of-Sale system. Needs a fast, intuitive interface optimized for speed above all else. Can apply discounts within authorized limits, process returns, and handle multiple payment methods (cash, card, check). Access is limited to POS-related functions.

5.5 Inventory Employee

Responsible for receiving products from suppliers, stock adjustments, inventory counting, transfers between locations, and warehouse organization. Works primarily with barcode scanners and mobile devices. Access is limited to inventory and warehouse functions.

5.6 Accountant

Uses financial reports, invoices, tax calculations, and accounting exports. Needs accurate, legally compliant invoicing and the ability to export data for official tax filings. Access is focused on financial modules and reports.

5.7 Supplier (Future)

A future portal will allow suppliers to receive purchase orders, confirm availability, update delivery status, and view payment history. This is planned for a future version and is not included in V1.

5.8 Customer (Future)

A future loyalty application will allow customers to view loyalty points, active promotions, purchase history, and receive personalized offers. This is planned for a future version.

Chapitre 6

Project Scope

6.1 Version 1.0 — Included Modules

Module	Description
Authentication	Secure login, logout, password reset, session management, RBAC
Multi-tenant Architecture	Row-Level Security, tenant isolation, tenant settings
Product Management	Catalog, categories, brands, barcodes, units, variants, pricing
Inventory Management	Stock tracking, movements, adjustments, counting, transfers, alerts
Warehouse Management	Locations, zones, shelves, bin management, transfer optimization
Point of Sale	Fast checkout, cart, discounts, holds, multiple payments, offline mode
Purchasing	Purchase orders, supplier quotes, delivery tracking, auto-reorder suggestions
Supplier Management	Supplier directory, evaluation, MCDA ranking, communication history
Customer Management	Customer records, loyalty points, purchase history, segments
Financial Management	Invoicing (DÉCRET 05-468), expenses, taxes (TVA), profit and loss
Employee Management	Staff records, roles, schedules, attendance, commissions
Dashboards	KPI overview, sales trends, inventory health, financial snapshots
Reporting	Sales, inventory, purchase, financial, employee reports with export
AI and Decision Support	Demand forecasting, supplier ranking, inventory optimization, waste prediction
Optimization Engine	AHP, TOPSIS, PROMETHEE for supplier evaluation and purchase decisions

6.2 Future Versions

Future releases may include : mobile applications, e-commerce integration, multi-vendor marketplace, delivery optimization, IoT integration (electronic scales, RFID, electronic shelf labels), computer vision for shelf monitoring, voice assistant for natural language queries, store layout optimization, queue optimization, and autonomous inventory counting.

6.3 Explicitly Out of Scope (V1)

Hardware manufacturing or sales, payment gateway integration (beyond basic cash and card recording), government tax filing API integration (manual export only), multi-currency support (DZD only for V1), biometric authentication, and blockchain or smart contract implementations.

Part II — Requirements Engineering

Chapitre 7

Functional Requirements

7.1 Authentication and Authorization (FR-AUTH)

ID	Requirement
FR-AUTH-01	The system shall allow users to authenticate using email and password.
FR-AUTH-02	The system shall securely hash passwords using bcrypt with a minimum cost factor of 12.
FR-AUTH-03	The system shall support secure logout that invalidates the session.
FR-AUTH-04	The system shall provide password reset functionality via email verification.
FR-AUTH-05	The system shall implement Role-Based Access Control (RBAC) with granular permissions.
FR-AUTH-06	The system shall lock user accounts after 5 consecutive failed login attempts for 15 minutes.
FR-AUTH-07	The system shall log all authentication events (login, logout, failed attempts) for audit purposes.
FR-AUTH-08	The system shall support multi-store access (a user may operate in multiple stores within a tenant).

7.2 Product Management (FR-PROD)

ID	Requirement
FR-PROD-01	The system shall support full CRUD operations for products (create, read, update, soft delete).
FR-PROD-02	The system shall support hierarchical product categories (unlimited depth, adjacency list model).
FR-PROD-03	The system shall support brand management with CRUD operations.
FR-PROD-04	The system shall generate barcodes in EAN-13 and Code-128 formats.
FR-PROD-05	The system shall support multiple barcodes per product, with one designated as primary.
FR-PROD-06	The system shall store three price levels : cost price, selling price, and wholesale price.
FR-PROD-07	The system shall track batch and lot numbers with expiration dates.
FR-PROD-08	The system shall support product search by name, barcode, category, and brand.
FR-PROD-09	The system shall support product import from CSV and Excel files.
FR-PROD-10	The system shall maintain price history (versioned pricing).

7.3 Inventory Management (FR-INV)

ID	Requirement
FR-INV-01	The system shall track real-time stock levels per product per location.
FR-INV-02	The system shall maintain an immutable log of all stock movements (every IN, OUT, and ADJUSTMENT).
FR-INV-03	The system shall automatically update stock levels via database triggers (never via application logic).
FR-INV-04	The system shall prevent stock from going below zero at the database level.
FR-INV-05	The system shall generate alerts when stock falls below the configurable minimum level.
FR-INV-06	The system shall track product expiration dates and generate alerts at 30, 15, and 7 days before expiry.
FR-INV-07	The system shall support full and partial inventory counting sessions.
FR-INV-08	The system shall require manager approval for inventory count adjustments.
FR-INV-09	The system shall support stock transfers between warehouses and stores.
FR-INV-10	The system shall support multiple stock valuation methods (FIFO, weighted average).

7.4 Point of Sale (FR-POS)

ID	Requirement
FR-POS-01	The system shall process a sale transaction in less than 500 milliseconds.
FR-POS-02	The system shall support product lookup via barcode scanning, name search, and category browsing.
FR-POS-03	The system shall support per-line and per-ticket discounts.
FR-POS-04	The system shall support multiple payment methods per transaction (cash, card, check, mixed).
FR-POS-05	The system shall support transaction hold and recall.
FR-POS-06	The system shall generate compliant receipts for thermal printers.
FR-POS-07	The system shall generate invoices compliant with DÉCRET 05-468 when requested by the customer.
FR-POS-08	The system shall function offline and queue transactions for synchronization when online.
FR-POS-09	The system shall support cash drawer management (open, close, audit).
FR-POS-10	The system shall generate end-of-shift X and Z reports.
FR-POS-11	The system shall support keyboard shortcuts for all primary operations.
FR-POS-12	The system shall process returns and refunds.

7.5 Purchasing (FR-PUR)

ID	Requirement
FR-PUR-01	The system shall support purchase order creation with full lifecycle (draft, approval, ordering, delivery, closure).
FR-PUR-02	The system shall require manager approval for purchase orders above a configurable threshold.
FR-PUR-03	The system shall support partial deliveries.
FR-PUR-04	The system shall record batch numbers and expiration dates upon delivery receipt.
FR-PUR-05	The system shall support supplier quotation comparison.
FR-PUR-06	The system shall generate automatic reorder suggestions based on stock levels and demand forecasts.

7.6 Financial Management (FR-FIN)

ID	Requirement
FR-FIN-01	The system shall generate invoices containing all mandatory fields per DÉCRET 05-468.
FR-FIN-02	The system shall enforce sequential invoice numbering with no gaps (database-level enforcement).
FR-FIN-03	The system shall calculate TVA at rates of 19% (standard), 9% (reduced), and 0% (exempt).
FR-FIN-04	The system shall calculate the tax stamp (droit de timbre) as 1% of TTC with a minimum of 100 DZD.
FR-FIN-05	The system shall generate invoices in PDF format with amount in words in French.
FR-FIN-06	The system shall store all tenant fiscal identifiers (NIF, NIS, RC, AI, legal form).
FR-FIN-07	The system shall track expenses by category and support expense reporting.
FR-FIN-08	The system shall generate a simplified profit and loss statement.
FR-FIN-09	The system shall generate TVA collected vs. TVA paid summaries for tax declaration.
FR-FIN-10	The system shall display all monetary values in DZD with French number formatting.

7.7 AI and Decision Support (FR-AI)

ID	**Requirement**
FR-AI-01	The system shall forecast demand per product per store using historical sales data.
FR-AI-02	The system shall incorporate seasonality and events (Ramadan, Eid) into demand forecasts.
FR-AI-03	The system shall rank suppliers using AHP for criteria weighting and TOPSIS for ranking.
FR-AI-04	The system shall support PROMETHEE as an alternative supplier ranking method.
FR-AI-05	The system shall calculate optimal reorder points and safety stock levels.
FR-AI-06	The system shall predict which products are at risk of expiring before sale.
FR-AI-07	The system shall recommend optimal purchase quantities based on forecasts and supplier scores.
FR-AI-08	The system shall support what-if scenario simulation (price changes, demand shifts).
FR-AI-09	The system shall display forecast confidence intervals (95% CI).
FR-AI-10	The system shall gracefully degrade when AI services are unavailable.

Chapitre 8

Non-Functional Requirements

8.1 Performance (NFR-PERF)

ID	Requirement
NFR-PERF-01	Average page load time shall remain below 2 seconds (P95).
NFR-PERF-02	POS transaction processing shall complete in under 500 milliseconds.
NFR-PERF-03	API response time shall be under 300ms for read operations and under 1s for write operations.
NFR-PERF-04	Dashboard data loading shall complete in under 3 seconds.

8.2 Scalability (NFR-SCALE)

ID	Requirement
NFR-SCALE-01	The multi-tenant architecture shall support thousands of businesses.
NFR-SCALE-02	The database shall handle over 100 million rows efficiently.
NFR-SCALE-03	The API layer shall support horizontal scaling.
NFR-SCALE-04	Static assets shall be served via CDN.

8.3 Availability (NFR-AVAIL)

ID	Requirement
NFR-AVAIL-01	Target availability shall be 99.9%.
NFR-AVAIL-02	The POS shall function offline (queue and sync).
NFR-AVAIL-03	The system shall degrade gracefully when AI services are unavailable.

8.4 Security (NFR-SEC)

ID	Requirement
NFR-SEC-01	Row-Level Security (RLS) shall provide database-level tenant isolation.
NFR-SEC-02	Sensitive data at rest shall be encrypted with AES-256.
NFR-SEC-03	Data in transit shall be encrypted with TLS 1.3.
NFR-SEC-04	Authentication shall use bcrypt with a minimum cost factor of 12.
NFR-SEC-05	RBAC shall enforce granular permissions at the service layer.
NFR-SEC-06	All mutations shall be logged in an immutable audit trail.
NFR-SEC-07	All API endpoints shall implement rate limiting.
NFR-SEC-08	CSRF protection shall be enabled on all state-changing endpoints.

8.5 Localization (NFR-I18N)

ID	Requirement
NFR-I18N-01	The system shall support French (primary), Arabic (RTL), and English.
NFR-I18N-02	All monetary values shall use DZD with French number formatting (1 250,00 DA).
NFR-I18N-03	Dates shall be formatted as DD/MM/YYYY.
NFR-I18N-04	The Arabic interface shall support full right-to-left layout.

Part III — Software Architecture

Chapitre 9

System Architecture

9.1 Architectural Overview

RetailOS follows a **monolithic frontend with modular backend** architecture. The frontend is a single Next.js application. The backend uses Next.js API Routes for CRUD operations and a separate Python microservice for AI and optimization workloads.

This approach was chosen because it simplifies deployment for the initial Algerian market (where DevOps expertise is limited), avoids the operational complexity of microservices for V1, and isolates the AI engine (which benefits from a Python runtime for ML libraries). The architecture can evolve toward microservices as the user base grows.

9.2 Technology Stack

9.3 Multi-Tenancy Strategy

RetailOS uses **PostgreSQL Row-Level Security (RLS)** for tenant isolation. All business tables contain a `tenant_id` column (UUID, NOT NULL, indexed). PostgreSQL RLS policies ensure that every query automatically filters by the authenticated tenant's ID.

The application sets a PostgreSQL session variable `app.current_tenant_id` at the start of each authenticated request via Prisma middleware. RLS policies reference this variable to filter rows :

```
CREATE POLICY tenant_isolation ON products
  USING (tenant_id = current_setting('app.current_tenant_id')::
     uuid);
```

This provides database-level isolation — even if application code has a bug, data from one tenant cannot leak to another.

9.4 Layered Architecture

Each module follows a consistent layered architecture :

1. **API Routes** — HTTP endpoint handlers with Zod validation.
2. **Services** — Business logic, authorization checks, orchestration.
3. **Repositories** — Data access via Prisma, database queries.
4. **Components** — React UI components (list, form, detail, delete).
5. **Hooks** — TanStack Query hooks for data fetching and mutations.
6. **Validators** — Zod schemas matching Prisma model shapes.

7. **Types** — TypeScript interfaces and type definitions.

Layer	**Technology**	**Justification**
Frontend	Next.js 16, React, TypeScript	App Router, SSR, API routes in one framework
Styling	Tailwind CSS 4, shadcn/ui	Rapid UI development, consistent design
State	Zustand, TanStack Query	Client state and server state separation
Forms	React Hook Form, Zod	Type-safe form validation
Backend	Next.js API Routes	Same framework, shared types
ORM	Prisma 6.x	Type-safe database access, migrations
Database	PostgreSQL 16+	RLS, JSONB, extensions, reliability
Cache	Redis	Session store, job queues, caching
Auth	NextAuth.js v5 (Auth.js)	JWT, session management, providers
Real-time	Socket.io	WebSocket for notifications and alerts
Jobs	BullMQ	Background job processing
AI Engine	Python, FastAPI	ML ecosystem (scikit-learn, Prophet, OR-Tools)
Storage	S3-compatible (MinIO)	Files, PDFs, images, exports

Table 9.1 – Technology Stack

Chapitre 10

Database Design

10.1 Design Principles

- Every business table has `tenant_id` (UUID, NOT NULL, indexed).
- Primary keys are UUID v4, generated by `uuid_generate_v4()`.
- Timestamps are mandatory : `created_at`, `updated_at`, `deleted_at` (soft delete).
- Audit trail columns : `created_by`, `updated_by` (UUID, references users).
- Monetary values are stored as `Int` in centimes (1 DZD = 100 centimes).
- Stock levels are updated via database triggers on `stock_movements`.
- Invoice numbers are sequential per tenant, enforced by a database sequence.

10.2 Entity-Relationship Overview

The database contains approximately 40 tables organized into the following domains :

10.3 Invoice Numbering

Invoice numbers follow the format `YYYY-NNNNN` (year + 5-digit sequential number). Sequentiality with no gaps is enforced at the database level using a dedicated sequence table :

```
CREATE TABLE invoice_sequences (
  tenant_id UUID PRIMARY KEY REFERENCES tenants(id),
  last_number INT NOT NULL DEFAULT 0,
  year INT NOT NULL DEFAULT EXTRACT(YEAR FROM CURRENT_DATE),
  UNIQUE (tenant_id, year)
);
```

10.4 Tax Stamp Calculation

The Algerian *droit de timbre* is calculated as follows :

```
tax_stamp = GREATEST(100, ROUND(total_ttc * 0.01));
net_to_pay = total_ttc + tax_stamp;
```

All values are in centimes. The minimum tax stamp is 100 DZD (10 000 centimes).

Domain	Tables
System	tenants, users, roles, permissions, role_permissions, user_roles, user_stores, stores
Products	products, product_categories, brands, units, product_barcodes, product_batches
Inventory	stock_movements, stock_levels, stock_transfers, stock_transfer_items, stock_counts, stock_count_items
Sales/POS	sales, sale_items, sale_payments, sale_returns, sale_return_items, pos_sessions, pos_cash_movements
Purchasing	purchase_orders, purchase_order_items, supplier_quotes, supplier_quote_items, purchase_deliveries, purchase_delivery_items, suppliers, supplier_contacts, supplier_products
Finance	invoices, invoice_items, invoice_payments, expenses, expense_categories, tax_rates, payment_methods, financial_periods
Customers	customers, loyalty_point_transactions, customer_segments, customer_debts, customer_debt_payments
Employees	employees, work_schedules, attendance_records, commissions, commission_rules
AI	supplier_evaluations, demand_forecasts, ai_recommendations
Audit	audit_logs

Table 10.1 – Database Table Domains

Part IV — AI and Optimization Engine

Chapitre 11

Decision Support Algorithms

11.1 Overview

The AI and Optimization Engine is the core differentiator of RetailOS. It transforms operational data into actionable business recommendations using established operations research and machine learning methodologies.

11.2 Demand Forecasting

11.2.1 Methodology

Demand forecasting uses a combination of statistical and machine learning models :

- **Prophet** (Facebook/Meta) — Decomposable time series model that handles seasonality, holidays, and trend changes. Particularly effective for retail data with weekly and yearly patterns.
- **ARIMA** — Classical time series model for stationary data. Used as a baseline and for products with limited history.
- **XGBoost** — Gradient boosting model that can capture complex non-linear relationships between product features, promotions, and demand.

11.2.2 Algerian Context

The forecasting model incorporates Algerian-specific events :

- **Ramadan** — The most significant demand event. Demand for food products increases by 30–80% in the weeks before and during Ramadan. The model uses the Islamic calendar (Hijri) to predict Ramadan dates.
- **Eid al-Fitr and Eid al-Adha** — Major demand spikes, especially for meat, sweets, and gifts.
- **Back-to-school (September)** — Increased demand for school supplies, snacks, and beverages.
- **National holidays** — Algerian Revolution Day (November 1), Independence Day (July 5), and other official holidays.
- **Seasonal patterns** — Summer demand for beverages and fresh products, winter demand for heating-related items.

11.2.3 Output

For each product and store, the model produces :

- Predicted daily demand (units)

— 95% confidence interval (lower and upper bounds)
— Model accuracy (MAPE — Mean Absolute Percentage Error)
— Detected trend (increasing, stable, decreasing)
— Detected seasonality (weekly, monthly, yearly)

11.3 Supplier Ranking using MCDA

11.3.1 Problem Definition

Supplier selection is a Multi-Criteria Decision Analysis (MCDA) problem. A supplier cannot be evaluated on price alone — delivery time, quality, reliability, payment terms, and product range all matter. The goal is to provide a systematic, reproducible ranking.

11.3.2 Stage 1 : AHP (Analytic Hierarchy Process)

The AHP method determines the relative weights of evaluation criteria through pairwise comparisons :

1. Define the criteria hierarchy :
   — Level 1 : Best Supplier
   — Level 2 : Price, Quality, Delivery Time, Reliability, Payment Terms, Product Range
2. Construct pairwise comparison matrices. For each pair of criteria, the decision maker indicates which is more important and by how much (on a scale of 1 to 9).
3. Calculate the priority vector (eigenvector) of each comparison matrix.
4. Check the Consistency Ratio (CR). If CR < 0.10, the comparisons are acceptably consistent. Otherwise, the decision maker should revise their judgments.

The mathematical formulation :

$$A \cdot w = \lambda_{max} \cdot w \tag{11.1}$$

Where A is the comparison matrix, w is the priority vector, and λ_{max} is the largest eigenvalue.

$$CI = \frac{\lambda_{max} - n}{n - 1} \tag{11.2}$$

$$CR = \frac{CI}{RI} \tag{11.3}$$

Where RI is the Random Index (depends on matrix size n).

11.3.3 Stage 2 : TOPSIS

TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) ranks suppliers based on their distance to the ideal and anti-ideal solutions :

1. Construct the decision matrix $D = [d_{ij}]$ where i = suppliers, j = criteria.
2. Normalize the matrix using vector normalization.
3. Apply the AHP weights from Stage 1 to obtain the weighted normalized matrix.
4. Determine the Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS).
5. Calculate the Euclidean distance of each supplier to PIS and NIS.
6. Calculate the closeness coefficient :

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-} \tag{11.4}$$

   Where d_i^+ is the distance to PIS and d_i^- is the distance to NIS.
7. Rank suppliers by CC_i (highest = best).

11.3.4 Alternative : PROMETHEE

PROMETHEE (Preference Ranking Organization METHod for Enrichment Evaluations) is used when criteria have non-linear preference functions. For example, delivery time may have a threshold : 1–2 days is ideal, 3–5 days is acceptable, and more than 5 days is unacceptable. PROMETHEE handles this through preference functions :

— **Type I : Usual criterion** — No preference threshold.
— **Type II : U-shape** — Indifference threshold only.
— **Type V : Linear with indifference and preference zone** — Most common for supplier evaluation.

PROMETHEE II produces a complete ranking with net outranking flows :

$$\phi(a) = \phi^+(a) - \phi^-(a) \tag{11.5}$$

Where $\phi^+(a)$ measures how much a outranks all other alternatives, and $\phi^-(a)$ measures how much all other alternatives outrank a.

11.4 Inventory Optimization

11.4.1 Economic Order Quantity (EOQ)

The classic EOQ formula determines the optimal order quantity that minimizes total inventory costs :

$$Q^* = \sqrt{\frac{2 \cdot D \cdot S}{H}} \tag{11.6}$$

Where :

— D = annual demand (units)
— S = ordering cost per order (DZD)
— H = holding cost per unit per year (DZD)

11.4.2 Reorder Point

The reorder point determines when to place a new order :

$$ROP = \bar{d} \cdot L + z_\alpha \cdot \sigma_d \cdot \sqrt{L} \tag{11.7}$$

Where :
— $\bar{d}$ = average daily demand
— L = lead time (days)
— z_α = safety factor (e.g., 1.65 for 95% service level)
— σ_d = standard deviation of daily demand

11.4.3 Safety Stock

Safety stock provides a buffer against demand and supply variability :

$$SS = z_\alpha \cdot \sqrt{L \cdot \sigma_d^2 + \bar{d}^2 \cdot \sigma_L^2} \tag{11.8}$$

Where σ_L is the standard deviation of lead time.

11.5 Waste Prediction

Products at risk of expiration are identified using a probability model :

$$P(\text{expiration}) = 1 - P(\text{sale before expiry}) \tag{11.9}$$

The probability of selling a product before its expiration date is estimated from historical sales velocity and remaining shelf life. Products with high expiration probability trigger recommendations : markdown, promotion, or transfer to a higher-traffic location.

Part V — Algerian Legal and Fiscal Compliance

Chapitre 12

Fiscal Compliance

12.1 Business Identification

Every Algerian business must have the following identifiers, all of which are stored and displayed on invoices :

12.2 Taxation

12.2.1 TVA (VAT)

The standard TVA rate in Algeria is 19%. A reduced rate of 9% applies to certain essential food products and services. A 0% rate applies to exempt categories.

The TVA calculation on an invoice :

```
tva_amount = SUM(line.amount_ht * line.tva_rate / 100)
total_ttc = SUM(line.amount_ht) + tva_amount
```

12.2.2 Tax Stamp (Droit de Timbre)

```
tax_stamp = GREATEST(100, ROUND(total_ttc * 0.01))
net_to_pay = total_ttc + tax_stamp
```

12.3 Invoice Requirements (DÉCRET EXÉCUTIF 05-468)

Every invoice must contain the following mandatory fields :

1. Sequential invoice number (no gaps allowed)
2. Date and place of issue
3. Seller's identification : NIF, NIS, RC, AI, legal name, address
4. Buyer's identification : name, address, NIF (if applicable)
5. Description of goods or services
6. Quantity
7. Unit price (in DZD)
8. Total amount before tax (HT)
9. TVA rate(s) applied and corresponding amount(s)
10. Tax stamp amount
11. Total amount including tax (TTC)
12. Payment terms and conditions

12.4 Legal Archive Period

All invoices and financial documents must be archived for a minimum of **10 years** per Algerian law. The system must support long-term document retention and retrieval.

12.5 Currency

The only legal tender for domestic transactions is the **Algerian Dinar (DZD)**. The display format follows French conventions :

— Space as thousands separator : 1 250 000

— Comma as decimal separator : 150,50

— Suffix : DA or DZD

— Example : 1 250,00 DA

All monetary values are stored as integers in centimes (1 DZD = 100 centimes) to avoid floating-point arithmetic errors.

Identifier	**Full Name**	**Description**
NIF	Numero d'Identification Fiscale	15-character alphanumeric tax ID issued by DGI
NIS	Numero d'Identification Statistique	Statistical ID issued by ONS
RC	Registre de Commerce	Commercial registration number issued by CNRC
AI	Article d'Imposition	Tax article number
Forme Juridique	Legal Form	SARL, SPA, EURL, EIRL, SNC, etc.

Table 12.1 – Algerian Business Identifiers

Part VI — Deployment and Infrastructure

Chapitre 13

Deployment Architecture

13.1 Development Environment

The development environment uses Docker Compose :

13.2 Production Architecture

The production architecture supports horizontal scaling :

- **Load Balancer :** Nginx or Caddy handles TLS termination, static asset serving, and request distribution.
- **Application Servers :** Multiple Next.js instances behind the load balancer.
- **Database :** PostgreSQL with primary for writes and read replicas for reporting.
- **Cache :** Redis for session storage, BullMQ job queues, and query caching.
- **AI Engine :** FastAPI service with access to a read replica of the database.
- **Object Storage :** S3-compatible storage (MinIO for self-hosted, or cloud provider) for files, PDFs, and exports.

13.3 Backup Strategy

- **Database :** Daily full backup + continuous WAL archiving. Point-in-time recovery capability.
- **Object Storage :** Cross-region replication (if using cloud provider).
- **Retention :** 30 days for daily backups, 12 months for monthly backups.
- **Recovery Time Objective (RTO) :** 4 hours.
- **Recovery Point Objective (RPO) :** 1 hour.

Service	**Port**	**Description**
Next.js App	3000	Application server
PostgreSQL	5432	Primary database
Redis	6379	Cache and job queues
FastAPI AI Engine	8000	Python AI/optimization service
MinIO	9000	S3-compatible object storage

Table 13.1 – Development Docker Services

Annexe A

Glossary

Term	Definition
AHP	Analytic Hierarchy Process — A multi-criteria decision-making method for determining criteria weights through pairwise comparisons.
CNRC	Centre National du Registre du Commerce — The Algerian commercial registration authority.
DGI	Direction Générale des Impôts — The Algerian tax authority.
DZD	Algerian Dinar — The official currency of Algeria.
EOQ	Economic Order Quantity — The optimal order quantity that minimizes total inventory costs.
FEFO	First Expired, First Out — An inventory management method prioritizing products closest to expiration.
FIFO	First In, First Out — An inventory valuation method assuming the oldest items are sold first.
MCDA	Multi-Criteria Decision Analysis — Methods for evaluating and ranking alternatives based on multiple criteria.
MOAD	Modélisation, Optimisation et Aide à la Décision — Modeling, Optimization, and Decision Support.
NIF	Numéro d'Identification Fiscale — A 15-character alphanumeric tax identification number.
NIS	Numéro d'Identification Statistique — A statistical identification number.
ONS	Office National des Statistiques — The Algerian national statistics office.
PROMETHEE	Preference Ranking Organization METHod for Enrichment Evaluations.
PROMETHEE	A multi-criteria ranking method using preference functions.
RLS	Row-Level Security — A PostgreSQL feature for row-level access control.
ROP	Reorder Point — The inventory level at which a new order should be placed.
SaaS	Software as a Service — A software delivery model where applications are hosted in the cloud.
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution — A multi-criteria ranking method.
TVA	Taxe sur la Valeur Ajoutée — Value Added Tax (19% standard rate in Algeria).

Annexe B

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